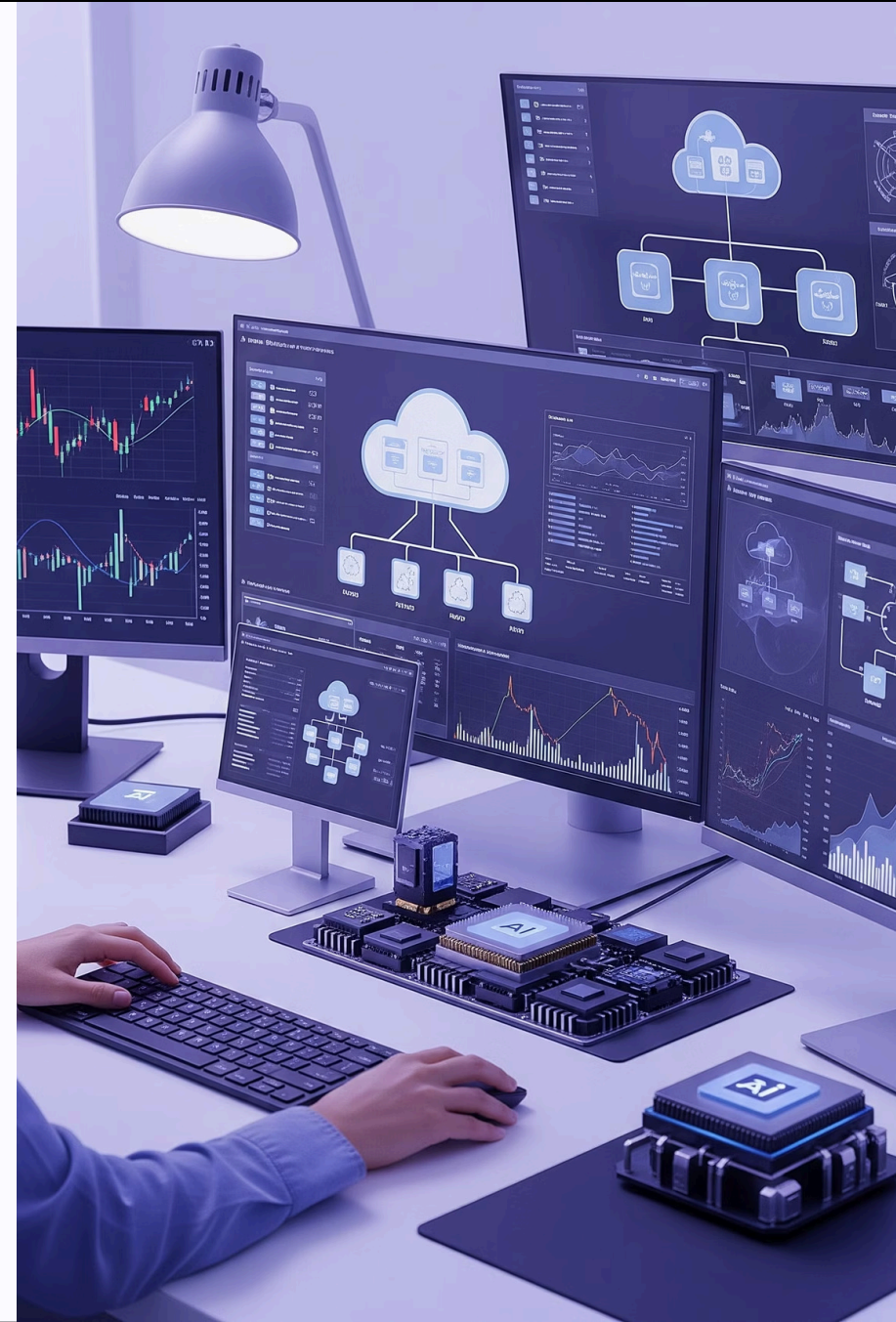


Estimating Hardware & Cloud Costs

A practical guide to budgeting AI infrastructure costs for your Capstone project



Why Cost Estimation Matters

Accurate cost estimation is the foundation of successful AI projects. Without it, you risk:

Resource Shortages

Prevents mid-project resource shortages that can derail your timeline

Informed Decisions

Guides critical **hardware vs cloud trade-offs** based on your project needs

Stakeholder Communication

Helps communicate technical feasibility to non-technical stakeholders

FinOps Culture

Supports a FinOps mindset for your Capstone project and future endeavors





compute

storage

networking

operations

Core Cost Categories

Understanding the four pillars of AI infrastructure costs

Compute

GPUs/TPUs/CPUs per hour

The processing power that drives model training and inference

Storage

Object, block, archive tiers

Where your datasets, checkpoints, and models live

Networking

Inter-region data transfer, API serving

Moving data between services and to users

Ops & Overhead

Monitoring, logging, compliance

The invisible but essential infrastructure

Compute Cost Models

On-premise Hardware

Upfront CAPEX + ongoing power/cooling costs

- Higher initial investment
- Lower cost over 3+ year timespan
- Complete control over hardware

\$3-4

NVIDIA A100 GPU

Per hour cloud rate

Cloud GPUs/TPUs

Pay-as-you-go hourly rates with discount options

- Flexibility to scale up/down
- No maintenance responsibilities
- Spot/preemptible instances for 70-90% discounts

\$6-8

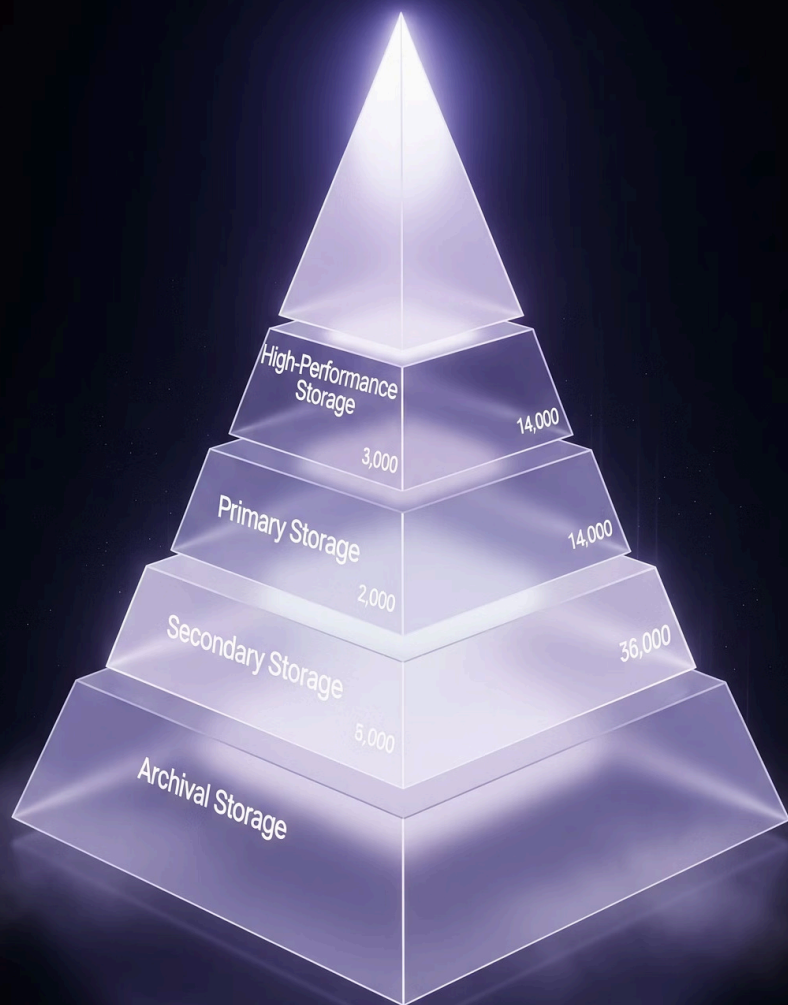
NVIDIA H100 GPU

Per hour cloud rate

\$2-3

TPU v5e

Per hour cloud rate



Storage Cost Models



Object Storage

~\$20–25/TB/month (S3/GCS/Azure Blob)

Standard for datasets, model artifacts, and general files



High-performance Block Storage

NVMe and high-IOPS solutions

2-5x cost of standard storage, but necessary for I/O-intensive workloads



Archival Storage

<\$5/TB/month

For rarely accessed data, with retrieval delays and fees



Data versioning multiplies storage costs if not carefully managed!

Networking Cost Models

Inter-region transfer is often the hidden cost driver in distributed AI workflows

Cloud Egress Fees

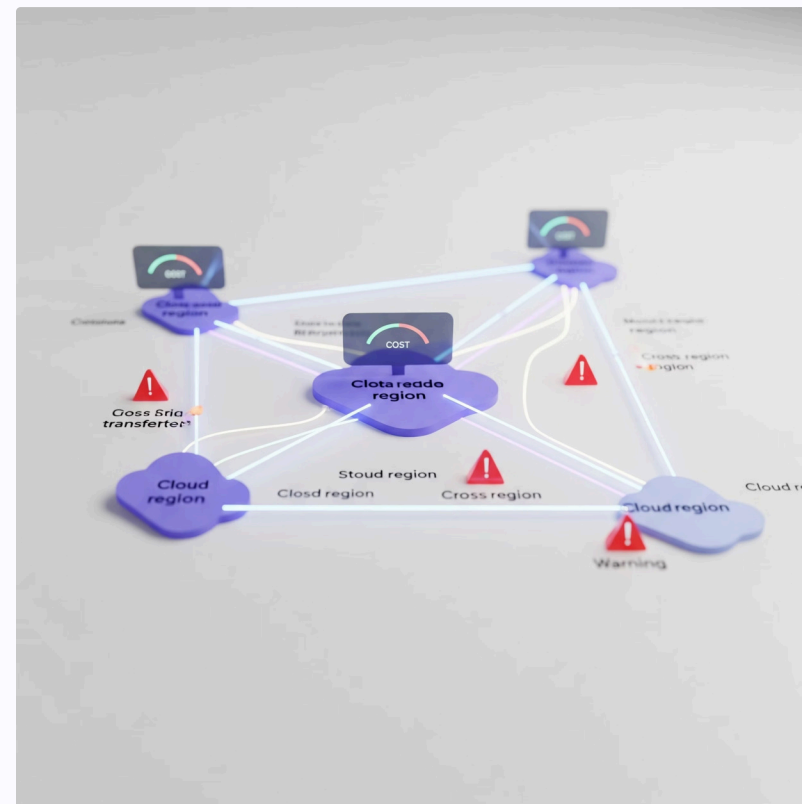
~\$0.05–0.12/GB to transfer data out of cloud providers

CDN Caching

Content delivery networks reduce inference serving costs by caching results closer to users

Co-location Strategy

Training is most cost-effective when **data + GPUs are co-located** in the same region



Hidden & Indirect Costs

The expenses that don't show up in the initial calculator

Logging & Monitoring

CloudWatch, Stackdriver, and similar services can add 5-15% to your total bill

Critical for debugging and performance optimization

Compliance Overhead

Audit storage, encryption, and compliance features add both direct costs and workflow complexity

Idle Resources

GPUs allocated but sitting unused are the #1 source of cloud waste

Automated shutdown policies are essential

Human Costs

DevOps, MLOps, and data labeling teams represent significant investments beyond infrastructure

Cost Optimization Strategies



Spot/Preemptible GPUs

Use interruptible instances for non-critical training jobs (70-90% savings)



Tiered Storage

Move data through hot → warm → cold tiers based on access patterns



Model Efficiency

Quantized and distilled models dramatically reduce inference costs



Right-size Resources

Don't overprovision clusters—match instance types to workload needs



- ✓ A well-optimized AI infrastructure can often deliver **40-60% cost savings** compared to default configurations

Case Study: NLP Capstone Cost Estimate

Training DistilBERT on Cloud GPUs

Resource	Specs	Cost
Compute	2× A100 GPUs for 48 hrs	~\$500
Storage	1 TB text corpus, 2 months	~\$50
Serving API	Small cluster, 1 month	~\$200
Total		≈ \$750

- ❏ NLP projects typically have smaller datasets but can require substantial serving capacity for real-time API access.



Case Study: Vision Capstone Cost Estimate

Training ResNet50 on Medical Image Dataset

Resource	Specs	Cost
Compute	4× A100 GPUs for 72 hrs	~\$1,200
Storage	5 TB medical images	~\$120
Inference	Edge + cloud deployment	~\$500
Total		≈ \$1,820



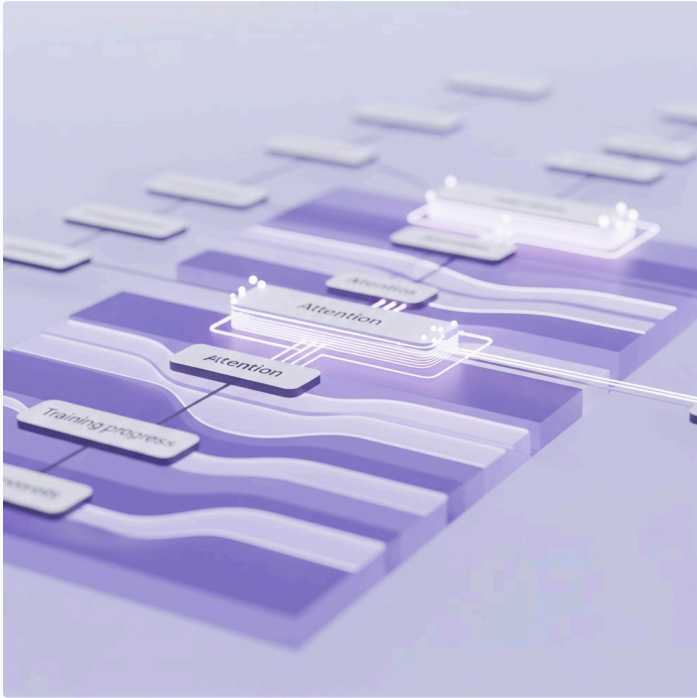
- ❑ Vision projects typically have higher storage requirements due to large image datasets, with compute costs scaling accordingly.

Case Study: GenAI Capstone Cost Estimate

Fine-tuning LLaMA 7B

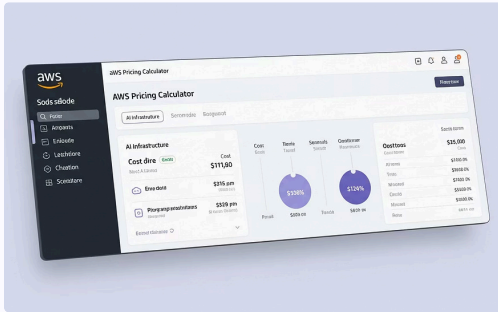
Resource	Specs	Cost
Compute	8× A100 GPUs for 120 hrs	~\$3,000
Storage	10 TB dataset	~\$200
Serving API	Autoscaling cluster, 2 months	~\$1,000
Total		≈ \$4,200

⚠ GenAI projects have the highest compute requirements, often necessitating multi-GPU setups for extended training periods.



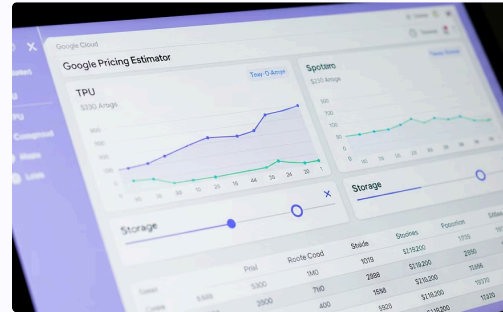
Tools for Cost Estimation

Leverage these platforms to create detailed cost projections



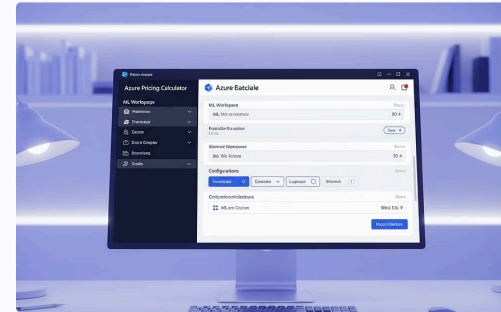
AWS Pricing Calculator

Comprehensive tool for estimating EC2, S3, and SageMaker costs



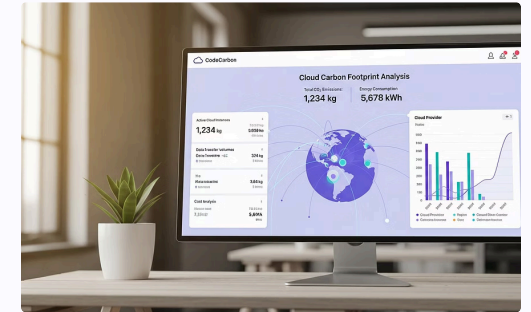
Google Cloud Pricing Estimator

Excellent for TPU-based workloads and Vertex AI pricing



Azure Pricing Calculator

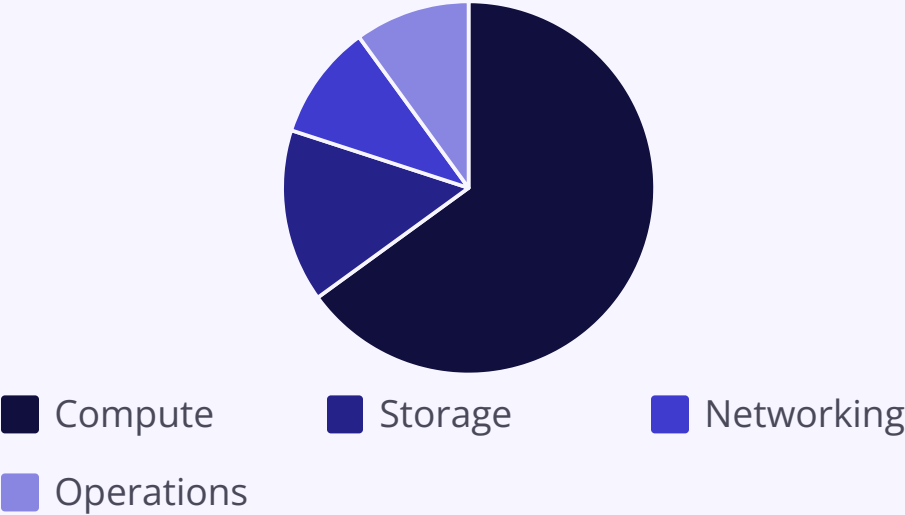
Strong for estimating Azure ML and cognitive services costs



CodeCarbon

Track both financial costs and CO₂ impact of your AI workloads

Cost Breakdown Structure



Typical AI project cost distribution

Layer	Cost Driver	Typical Cost
Compute	GPU hours	\$3–8/hr/GPU
Storage	Object TB/month	\$20–25
Networking	Data transfer/GB	\$0.05–0.12
Ops	Monitoring/logs	5–10% extra

"Cloud costs = compute-heavy, storage/network hidden drivers."

Key Lessons for Budget Planning

1 Compute Dominates, But Don't Ignore the Rest

While GPU/TPU costs will be your largest line item, storage and network transfer add up quickly, especially at scale

2 Use Cloud Calculators As Essential Planning Tools

Don't guess—leverage provider-specific calculators to create accurate projections before committing resources

3 Leverage Spot/Preemptible Instances

These can cut training costs by 70-90%, with proper checkpointing to manage interruptions

4 Always Budget 20–30% Overhead

Hidden costs will emerge—from logging to idle resources to unexpected scaling needs



Discussion Prompt

"If you had a \$1,000 Capstone budget, would you spend more on bigger GPUs for faster training or longer inference serving capacity — and why?"



Consider these factors in your response:

- Project type and audience requirements
- Iterative development needs
- Demo expectations for final presentation
- Long-term deployment goals
- Team size and division of responsibilities